Lecture 8: Cross-Filling Projections

MATH203 Applied Linear Algebra

Cross-Filling a Projection

Exercise: Compute the Rank

The following matrix is a projection ($P^2 = P$).

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

Task: Compute $\operatorname{rank}(P)$ by cross-filling.

Cross-Filling: Step 1

$P_{(1,1)} = 0$ — skip. **First pivot** at $(2,2)$, value $= \mathbf{1}$:

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & \mathbf{1} & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

$$R_1 = \frac{1}{\mathbf{1}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} -2 & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Cross-Filling: Step 1 — The Decomposition

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}}_{\text{remainder}}$$

A rank-1 piece, plus what's left. Standard cross-filling.

But let's look more carefully at these two pieces...

Wait — R_1 Is a Projection?

Let's check: is $R_1^2 = R_1$?

$$R_1^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R_1 \quad \checkmark$$

And the **remainder** $P - R_1$?

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} \quad \checkmark$$

The rank-1 piece **and** the remainder are **both projections**. Coincidence?

Cross-Filling: Step 2

Second pivot at $(3,3)$, value $= \mathbf{1}$ (in the remainder):

$$P - R_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & \mathbf{1} & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

$$R_2 = \frac{1}{\mathbf{1}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Cross-Filling: Step 2 — The Decomposition

$$P - R_1 = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{R_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}}_{\text{remainder}}$$

$R_2^2 = R_2$ ✓ Again a projection! And the remainder too.

Cross-Filling: Step 3

Third pivot at $(4,4)$, value $= \mathbf{1}$ (in the remainder):

$$P - R_1 - R_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & \mathbf{1} \end{pmatrix}$$

$$R_3 = \frac{1}{\mathbf{1}} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & \mathbf{1} \end{pmatrix}$$

$R_3^2 = R_3$ ✓ Every single piece is a projection.

What Just Happened?

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{R_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}}_{R_3}$$

Every piece is a projection: $R_1^2 = R_1, \quad R_2^2 = R_2, \quad R_3^2 = R_3$ ✓

Pieces annihilate each other: $R_i R_j = 0$ for $i \neq j$ ✓

Rank equals trace: $\text{rank}(P) = 3 = 0 + 1 + 1 + 1 = \text{trace}(P)$ ✓

For Contrast: A Non-Projection

Does this happen for **every** matrix? **No!**

$$A = \begin{pmatrix} \mathbf{2} & 4 \\ 1 & 2 \end{pmatrix} \quad (\text{rank } 1, \text{ but } A^2 \neq A)$$

Pivot at $(1, 1)$, value $= \mathbf{2}$:

$$R_1 = \frac{1}{\mathbf{2}} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{pmatrix} 2 & 4 \end{pmatrix} = \begin{pmatrix} \mathbf{2} & 4 \\ 1 & 2 \end{pmatrix}$$

Rank $= 1$, so $A = R_1$ (one piece, no remainder).

For Contrast: Everything Fails

Is R_1 a projection?

$$R_1^2 = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 16 \\ 4 & 8 \end{pmatrix} = \mathbf{4}\, R_1 \neq R_1 \quad \times$$

Does rank = trace?

$$\underbrace{\text{rank}(A) = 1}_{\text{one piece}} \qquad \underbrace{\text{trace}(A) = 2 + 2 = 4}_{\text{sum of diagonal}} \qquad 1 \neq 4 \quad \times$$

None of the nice properties hold — because A is **not a projection**.

Why does cross-filling a projection produce projections?

What is the hidden structure that makes $P^2 = P$ so special?

Why It Works: $P = UV$ Implies $VU = I$

Back to Our Example: $P = R_1 + R_2 + R_3$

Recall the cross-filling decomposition:

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{R_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}}_{R_3}$$

Each $R_i = \underbrace{\text{column} \div \text{pivot}}_{\mathbf{u}_i} \cdot \underbrace{\text{row}}_{\mathbf{v}_i^T}$. Can we collect all columns and rows?

Collecting into $P = UV$

Stack the columns into U, the rows into V:

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{U\,:\,\text{columns}\div\text{pivot}} \underbrace{\begin{pmatrix} -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}}_{V\,:\,\text{rows}}$$

$\mathbf{u}_1\mathbf{v}_1^T = R_1, \quad \mathbf{u}_2\mathbf{v}_2^T = R_2, \quad \mathbf{u}_3\mathbf{v}_3^T = R_3$

Compute VU

$$VU = \begin{pmatrix} -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3$$

$VU = I_3$ **automatically**!

Recall the non-projection: $VU = 4 \neq I_1$. So this is about $P^2 = P$.

Is $VU = I$ **always** true when P is a projection?

The Key Theorem

Theorem ($VU = I$ Automatic)

Let P be a projection ($P^2 = P$). Suppose $P = UV$ where:

- U is $n \times r$ with linearly independent columns
- V is $r \times n$ with linearly independent rows

Then $VU = I_r$ **automatically**.

This single identity explains **everything** we discovered in Act 1.

Proof: Three Lines

Expand $P^2 = P$ using $P = UV$:

$$\underbrace{UV \cdot UV}_{P^2} = \underbrace{UV}_{P}$$

Step 1: Left-cancel U (columns of U are linearly independent):

$$UVUV = UV \quad \Longrightarrow \quad VUV = V$$

Step 2: Right-cancel V (rows of V are linearly independent):

$$VUV = V \quad \Longrightarrow \quad VU = I_r$$

Done! The proof uses **only** $P^2 = P$ and cancellation. $\square$

Now We Can Explain Everything

Write $P = UV$ with $VU = I_r$:

$$U = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_r \\ | & | & & | \end{pmatrix}, \qquad V = \begin{pmatrix} - & \mathbf{v}_1^T & - \\ - & \mathbf{v}_2^T & - \\ & \vdots & \\ - & \mathbf{v}_r^T & - \end{pmatrix}$$

Each rank-1 piece: $R_1 = \mathbf{u}_1\mathbf{v}_1^T, \quad R_2 = \mathbf{u}_2\mathbf{v}_2^T, \quad \ldots \quad R_r = \mathbf{u}_r\mathbf{v}_r^T$

What Does $VU = I_r$ Look Like?

$$VU = \begin{pmatrix} \mathbf{v}_1^T\mathbf{u}_1 & \mathbf{v}_1^T\mathbf{u}_2 & \cdots & \mathbf{v}_1^T\mathbf{u}_r \\ \mathbf{v}_2^T\mathbf{u}_1 & \mathbf{v}_2^T\mathbf{u}_2 & \cdots & \mathbf{v}_2^T\mathbf{u}_r \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_r^T\mathbf{u}_1 & \mathbf{v}_r^T\mathbf{u}_2 & \cdots & \mathbf{v}_r^T\mathbf{u}_r \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = I_r$$

Diagonal: $\mathbf{v}_i^T\mathbf{u}_i = 1$ (same-color pairs multiply to 1)

Off-diagonal: $\mathbf{v}_i^T\mathbf{u}_j = 0$ for $i \neq j$ (different-color pairs multiply to 0)

Each Piece Is a Projection

Goal: Show $R_i^2 = R_i$ using $\mathbf{v}_i^T \mathbf{u}_i = 1$.

$$R_i^2 = (\mathbf{u}_i \mathbf{v}_i^T)(\mathbf{u}_i \mathbf{v}_i^T) = \mathbf{u}_i \underbrace{(\mathbf{v}_i^T \mathbf{u}_i)}_{=1} \mathbf{v}_i^T = \mathbf{u}_i \mathbf{v}_i^T = R_i \quad \checkmark$$

Each cross-filling piece $R_i = \mathbf{u}_i \mathbf{v}_i^T$ is **automatically a rank-1 projection**!

Mutually Annihilating Pieces

Goal: Show $R_i R_j = 0$ (for $i \neq j$) using $\mathbf{v}_i^T \mathbf{u}_j = 0$.

$$R_i R_j = (\mathbf{u}_i \mathbf{v}_i^T)(\mathbf{u}_j \mathbf{v}_j^T) = \mathbf{u}_i \underbrace{(\mathbf{v}_i^T \mathbf{u}_j)}_{=0} \mathbf{v}_j^T = 0 \quad \checkmark$$

Summary: For a projection $P = R_1 + \cdots + R_r$ (cross-filling decomposition):

- Each R_i is a rank-1 projection ($R_i^2 = R_i$)
- Different pieces annihilate ($R_i R_j = 0$ for $i \neq j$)

Both follow from the single identity $VU = I_r$.

Rank Equals Trace for Projections

Remember This?

Our projection from Act 1:

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

$$\text{trace}(P) = 0 + 1 + 1 + 1 = 3 = \text{rank}(P)$$

The diagonal entries sum to the rank. **Coincidence?**

Rank Equals Trace

Theorem (Rank = Trace for Projections)

For any projection P ($P^2 = P$):

$$\text{rank}(P) = \text{trace}(P)$$

Proof: Write $P = UV$ with $\text{rank}(P) = r$. By the previous theorem, $VU = I_r$.

$$\text{trace}(P) = \text{trace}(UV) = \text{trace}(VU) = \text{trace}(I_r) = r = \text{rank}(P)$$

Done — one line from $VU = I$. $\square$

The Two Extremes

What are the **smallest** and **largest** projections?

Proposition (The Smallest: $P = 0$)

If P is a projection with $\text{rank}(P) = 0$, then $P = 0$.

rank $= 0$ means every column is zero. ✓

Proposition (The Largest: $P = I$)

If P is an $n \times n$ projection with $\text{trace}(P) = n$, then $P = I_n$.

Proof: $I_n - P$ is a projection with $\text{trace}(I_n - P) = n - n = 0$.

By rank $=$ trace: $\text{rank}(I_n - P) = 0$, so $I_n - P = 0$. $\square$

Why Only for Projections?

Remember our non-projection from Act 1?

	rank	trace	Equal?
$\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$ (**Act 1**, not projection)	1	4	$\times$
$\begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}$ (not projection)	1	3	$\times$
$\frac{1}{5}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ (projection)	1	1	$\checkmark$
Our 4×4 example (**Act 1**, projection)	3	3	$\checkmark$

rank = trace is a **fingerprint** of projection matrices.

Quick Application: Read the Rank from the Diagonal

Exercise

The following matrix is a projection ($P^2 = P$). What is its rank?

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Answer: $\text{rank}(P) = \text{trace}(P) = 0 + 1 + 1 + 1 + 1 = 4$.

No cross-filling needed — just read the diagonal!

Orthogonal Projections: Orthonormal Bases for Free

What About Orthogonal Projections?

So far: cross-filling **any** projection gives rank-1 projections.

But an **orthogonal projection** carries **more structure**: $P^2 = P$ **and** $P^T = P$.

Our rank-1 pieces R_i inherited $R_i^2 = R_i$. Can they also inherit $R_i^T = R_i$?

Yes — if we choose pivots **on the diagonal**.

Let's see why.

Exercise: Cross-Fill Along the Diagonal

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \qquad P^2 = P \checkmark \qquad P^T = P \checkmark$$

Task: Cross-fill P using **diagonal pivots** only.

(Rank $=$ trace $= 1 + \frac{1}{2} + \frac{1}{2} = 2$, so we expect 2 steps.)

Diagonal Cross-Filling: Step 1

Pivot at $(\mathbf{1}, \mathbf{1})$, value $= \mathbf{1}$:

$$P = \begin{pmatrix} \mathbf{1} & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$R_1 = \frac{1}{\mathbf{1}} \underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}_{\text{column}} \cdot \underbrace{\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}}_{\text{row} = \text{column}^T!} = \begin{pmatrix} \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Step 1 — The Decomposition

$$P = \underbrace{\begin{pmatrix} \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_1 \text{ (symmetric } \checkmark)} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{\text{remainder (also symmetric } \checkmark)}$$

R_1 is a projection ($R_1^2 = R_1$ ✓) — as expected from §1.

But now R_1 is also **symmetric**: $R_1^T = R_1$ ✓.

Diagonal Cross-Filling: Step 2

Pivot at $(\mathbf{2}, \mathbf{2})$ in the remainder, value $= \frac{\mathbf{1}}{\mathbf{2}}$:

$$P - R_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{\mathbf{1}}{\mathbf{2}} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$R_2 = \frac{1}{\frac{\mathbf{1}}{\mathbf{2}}} \underbrace{\begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}}_{\text{column}} \cdot \underbrace{\begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{\text{row} = \text{column}^T!} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{\mathbf{1}}{\mathbf{2}} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

Again: row $=$ column^T, and R_2 is **symmetric** ✓

Row = Column^T — So What?

Since row = column^T, each piece factors as:

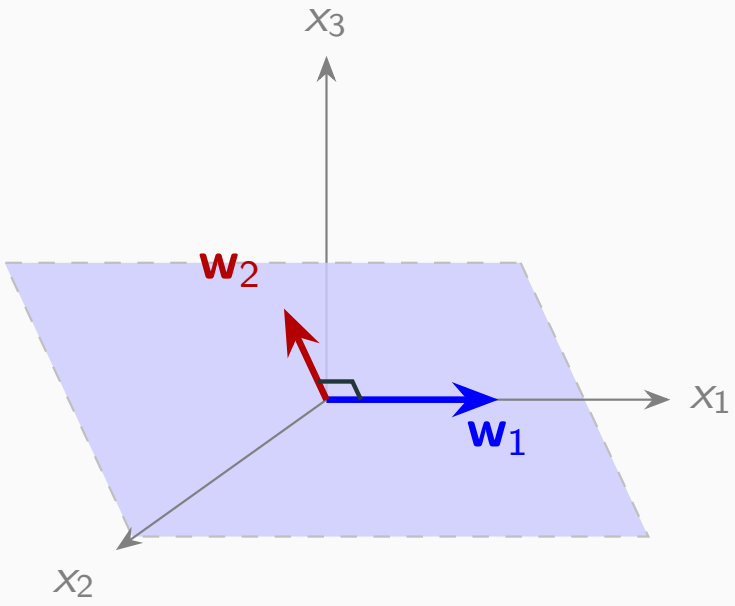
$$R_i = \frac{1}{\text{pivot}} \cdot \text{col} \cdot \text{col}^T = \underbrace{\frac{\text{col}}{\sqrt{\text{pivot}}}}_{\mathbf{w}_i} \cdot \underbrace{\frac{\text{col}^T}{\sqrt{\text{pivot}}}}_{\mathbf{w}_i^T} = \mathbf{w}_i \mathbf{w}_i^T$$

Step 1: $\text{col} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\text{pivot} = 1 \quad \Longrightarrow \quad \mathbf{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Step 2: $\text{col} = \begin{pmatrix} 0 \\ 1/2 \\ 1/2 \end{pmatrix}$, $\text{pivot} = \frac{1}{2} \quad \Longrightarrow \quad \mathbf{w}_2 = \frac{1}{\sqrt{1/2}} \begin{pmatrix} 0 \\ 1/2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

Let's Draw Them

$$P = \mathbf{w}_1\mathbf{w}_1^T + \mathbf{w}_2\mathbf{w}_2^T, \quad \text{where } \mathbf{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$



$$\|\mathbf{w}_1\| = 1, \quad \|\mathbf{w}_2\| = 1, \quad \mathbf{w}_1^T\mathbf{w}_2 = 0$$

Perpendicular. Unit length. **Is there a name for this?**

Recall: The Inner Product

The **inner product** (dot product) of two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{u}^T\mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Two key uses:

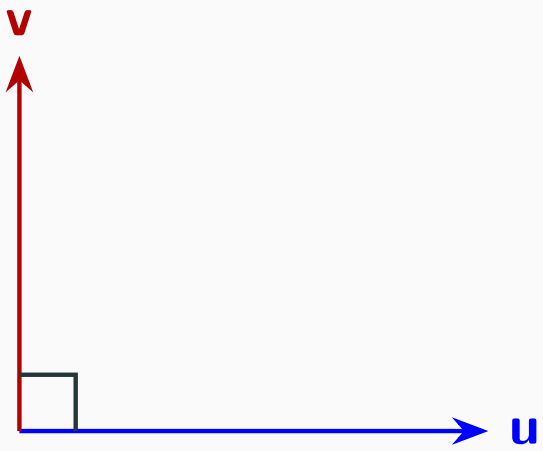
- **Length**: $\|\mathbf{u}\| = \sqrt{\mathbf{u}^T\mathbf{u}}$
- **Angle**: $\mathbf{u}^T\mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\| \cos\theta$

  In particular: $\mathbf{u}^T\mathbf{v} = 0 \iff \theta = 90$ (perpendicular!)

Orthogonal Vectors

Definition (Orthogonal)

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\mathbf{u}^T\mathbf{v} = 0$.



$$\mathbf{u}^T\mathbf{v} = 0$$

Orthogonal = perpendicular = "no overlap in direction."

Orthonormal Vectors

Definition (Orthonormal Set)

Vectors $\mathbf{w}_1, \ldots, \mathbf{w}_r$ are **orthonormal** if:

1. **Orthogonal**: $\mathbf{w}_i^T \mathbf{w}_j = 0$ for all $i \neq j$ (perpendicular pairs)
2. **Unit length**: $\|\mathbf{w}_i\| = 1$ for all i

Compact notation: $\mathbf{w}_i^T \mathbf{w}_j = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$

An **orthonormal basis** of a subspace S: an orthonormal set that also **spans** S.

Back to Our Vectors — They Are Orthonormal!

Recall: diagonal cross-filling gave us $P = \mathbf{w}_1\mathbf{w}_1^T + \mathbf{w}_2\mathbf{w}_2^T$ with

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{w}_2 = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

Unit length? $\|\mathbf{w}_1\|^2 = 1$ ✓ $\|\mathbf{w}_2\|^2 = \frac{1}{2} + \frac{1}{2} = 1$ ✓

Orthogonal? $\mathbf{w}_1^T\mathbf{w}_2 = 1 \cdot 0 + 0 \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{1}{\sqrt{2}} = 0$ ✓

Span $\mathrm{Col}(P)$**?** $P = \mathbf{w}_1\mathbf{w}_1^T + \mathbf{w}_2\mathbf{w}_2^T$ ✓ (every $P\mathbf{x}$ is a combination of $\mathbf{w}_1, \mathbf{w}_2$)

$\mathbf{w}_1, \mathbf{w}_2$ is an **orthonormal basis** of $\mathrm{Col}(P)$ — **for free**!

The Theorem

Theorem (Diagonal Cross-Filling of Orthogonal Projections)

Let P be an orthogonal projection ($P^2 = P$, $P^T = P$) of rank r.

Diagonal cross-filling decomposes P into:

$$P = \mathbf{w}_1\mathbf{w}_1^T + \mathbf{w}_2\mathbf{w}_2^T + \cdots + \mathbf{w}_r\mathbf{w}_r^T$$

where $\mathbf{w}_1, \ldots, \mathbf{w}_r$ is an **orthonormal basis** of $\mathrm{Col}(P)$.

We verified this for our 3×3 example. But **why** must it always work?

Step 1: Write $P = WW^T$

Recall: $\mathbf{w}_i = \text{column}_i / \sqrt{\text{pivot}_i}$. Stack them into W:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix}}_{W = (\,\mathbf{w}_1 \;\; \mathbf{w}_2\,)} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}}_{W^T = \begin{pmatrix} \mathbf{w}_1^T \\ \mathbf{w}_2^T \end{pmatrix}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{P}$$

$P = \mathbf{w}_1\mathbf{w}_1^T + \mathbf{w}_2\mathbf{w}_2^T$ — same W on **both** sides.

Step 2: $W^T W = I$

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}}_{W^T} \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix}}_{W} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{I_2} \quad \checkmark$$

Why must this hold? $P^2 = P$:

$$WW^T \cdot WW^T = WW^T \implies W(\mathbf{W^T W})W^T = W(\mathbf{I})W^T$$

Cancel W and W^T from both sides: $\mathbf{W^T W = I}$.

$W^T W = I$ — Spelled Out

$$\underbrace{\begin{pmatrix} \mathbf{w}_1^T \\ \mathbf{w}_2^T \end{pmatrix}}_{W^T} \underbrace{\begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 \end{pmatrix}}_{W} = \begin{pmatrix} \mathbf{w}_1^T \mathbf{w}_1 & \mathbf{w}_1^T \mathbf{w}_2 \\ \mathbf{w}_2^T \mathbf{w}_1 & \mathbf{w}_2^T \mathbf{w}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{I_2}$$

Diagonal: $\mathbf{w}_1^T \mathbf{w}_1 = 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 = 1$ ✓
$\mathbf{w}_2^T \mathbf{w}_2 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 1$ ✓

Off-diagonal: $\mathbf{w}_1^T \mathbf{w}_2 = 1 \cdot 0 + 0 \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{1}{\sqrt{2}} = 0$ ✓

$\mathbf{w}_1, \mathbf{w}_2$ is an **orthonormal basis** of $\mathrm{Col}(P)$ — **for free**! $\square$

Summary: $P^T = P \implies P = WW^T$; $\quad P^2 = P \implies W^T W = I$ (orthonormal).

$\mathrm{Col}(P)$ ✓ — **What About** $\mathrm{Null}(P)$**?**

We just showed: diagonal cross-fill an orthogonal projection P

$$\implies \quad \text{orthonormal basis of } \mathrm{Col}(P).$$

New question: Can we find an orthonormal basis of $\mathrm{Null}(P)$?

Exercise

$$P = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

P is an orthogonal projection ($P^2 = P$, $P^T = P$), $\text{rank}(P) = 1$.

Task: Find an orthonormal basis of $\text{Null}(P)$.

The Trick: $\mathrm{Null}(P) = \mathrm{Col}(I - P)$

$Q = I - P$ is also an orthogonal projection ($Q^2 = Q$, $Q^T = Q$), rank $= 3$.

$$Q = I - P = \begin{pmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

Cross-fill $Q \implies$ orthonormal basis of $\mathrm{Col}(Q) = \mathrm{Null}(P)$.

Cross-Fill Q: Step 1

Pivot at $(\mathbf{1}, \mathbf{1})$, value $= \frac{\mathbf{3}}{\mathbf{4}}$:

$$Q = \begin{pmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

$$\text{col} = \begin{pmatrix} 3/4 \\ -1/4 \\ -1/4 \\ -1/4 \end{pmatrix}, \quad \text{pivot} = \frac{3}{4} \qquad \Longrightarrow \qquad \mathbf{w}_1 = \frac{\text{col}}{\sqrt{3/4}} = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{6} \\ -\frac{\sqrt{3}}{6} \\ -\frac{\sqrt{3}}{6} \end{pmatrix}$$

Cross-Fill Q: Step 2

Pivot at $(\mathbf{2}, \mathbf{2})$ in the remainder, value $= \frac{\mathbf{2}}{\mathbf{3}}$:

$$Q - R_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

$$\text{col} = \begin{pmatrix} 0 \\ 2/3 \\ -1/3 \\ -1/3 \end{pmatrix}, \quad \text{pivot} = \frac{\mathbf{2}}{\mathbf{3}} \qquad \Longrightarrow \qquad \mathbf{w}_2 = \frac{\text{col}}{\sqrt{2/3}} = \begin{pmatrix} 0 \\ \frac{\sqrt{6}}{3} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \end{pmatrix}$$

Cross-Fill Q: Step 3

Pivot at $(\mathbf{3}, \mathbf{3})$ in the remainder, value $= \frac{\mathbf{1}}{\mathbf{2}}$:

$$Q - R_1 - R_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\mathbf{1}}{\mathbf{2}} & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$\text{col} = \begin{pmatrix} 0 \\ 0 \\ 1/2 \\ -1/2 \end{pmatrix}, \quad \text{pivot} = \frac{\mathbf{1}}{\mathbf{2}} \qquad \Longrightarrow \qquad \mathbf{w}_3 = \frac{\text{col}}{\sqrt{1/2}} = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$W^T W = I$ — Orthonormal Basis of $\mathrm{Null}(P)$

$$\underbrace{\begin{pmatrix} \mathbf{w}_1^T \\ \mathbf{w}_2^T \\ \mathbf{w}_3^T \end{pmatrix}}_{W^T} \underbrace{\begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 \end{pmatrix}}_{W} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{I_3} \quad \checkmark$$

Verify: $\|\mathbf{w}_1\|^2 = \frac{3}{4} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = 1 \ \checkmark$
$\mathbf{w}_1^T \mathbf{w}_2 = 0 - \frac{\sqrt{18}}{18} + \frac{\sqrt{18}}{36} + \frac{\sqrt{18}}{36} = 0 \ \checkmark$

$\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ is an **orthonormal basis** of $\mathrm{Null}(P)$ — **for free**!

What If $UV = I$?

The Reverse Question

In §2 we discovered:

$$P = UV, \quad P^2 = P \quad \Longrightarrow \quad VU = I_r$$

Projection $\longrightarrow$ identity.

Reverse: Suppose we **start** with $UV = I_n$.

What is VU?

VU Is Always a Projection

Theorem

If $UV = I_n$ (U is $n \times r$, V is $r \times n$), then $(VU)^2 = VU$.

Proof:

$$(VU)(VU) = V\underbrace{(UV)}_{=I_n}U = VU \qquad \square$$

The circle closes:

$$\text{Projection} \xrightarrow{\S 2} \text{Identity} \xrightarrow{\text{reverse}} \text{Projection}$$

The Rank Is Determined

VU is an $r \times r$ projection. By rank = trace (§3):

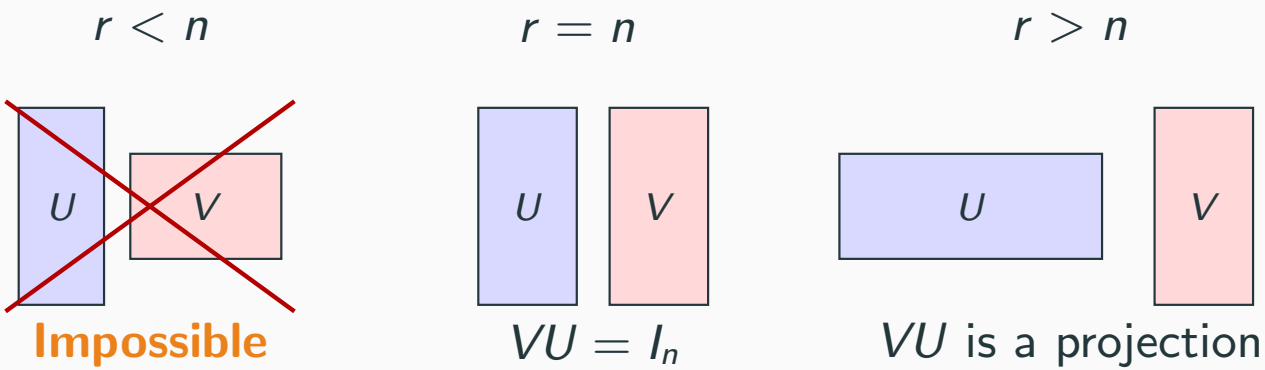
$$\text{rank}(VU) = \text{trace}(VU) = \text{trace}(UV) = \text{trace}(I_n) = \mathbf{n}$$

So VU is an $r \times r$ projection of rank n.

Since rank $\leq$ size: $\quad n \leq r$.

Three Cases by Shape

U is $n \times r$, $\quad V$ is $r \times n$, $\quad UV = I_n$, $\quad$ and $n \leq r$.



- $r < n$ (thin $\times$ fat): $\text{rank}(VU) \leq r < n$ — contradicts rank $= n$. **Impossible.**
- $r = n$ (square $\times$ square): rank-n projection on $\mathbb{R}^n \implies VU = I_n$.
- $r > n$ (fat $\times$ thin): VU is a **genuine** rank-n projection on $\mathbb{R}^r$.

The Square Case: One-Sided $\Rightarrow$ Two-Sided

Corollary

If U, V are both $n \times n$ and $UV = I_n$, then $VU = I_n$.

Proof: VU is an $n \times n$ projection of rank n (the square case).

A full-rank projection must be I (proved in Lecture 7; quick reminder if needed):

$I_n - VU$ is also a projection, and

$$\text{rank}(I_n - VU) = \text{trace}(I_n - VU) = n - n = 0 \implies VU = I_n. \qquad \square$$

Recall Lecture 1: We defined "invertible" by requiring **both** $AB = I$ and $BA = I$.

Now we know: for square matrices, **one equation suffices**!

$$UV = I_n \implies VU = I_n$$

Full Rank $\iff$ Invertible

When Is a Matrix Invertible?

§5 gave us a powerful tool:

$$UV = I_n \text{ (square)} \quad \Longrightarrow \quad VU = I_n$$

One-sided inverse $\Rightarrow$ two-sided. But **when** does a matrix have an inverse?

Answer: Exactly when it has **full rank**.

Full Rank $\iff$ Invertible

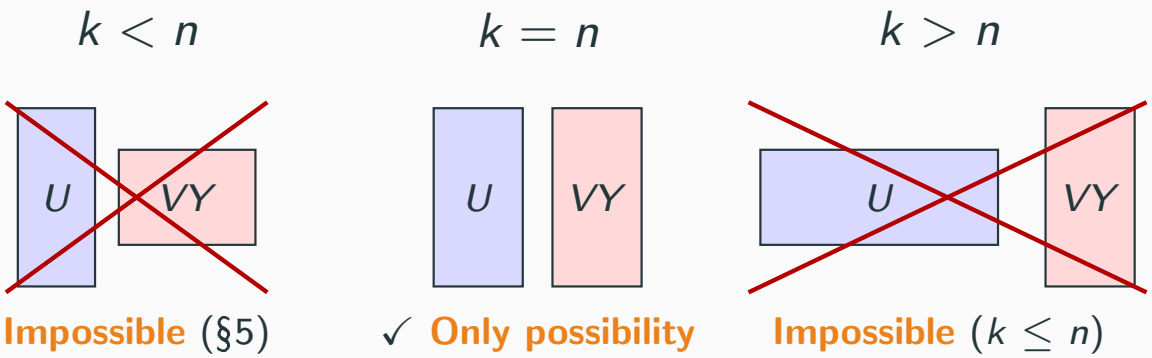
Theorem (Full Rank $\iff$ Invertible)

An $n \times n$ matrix X is invertible if and only if $\text{rank}(X) = n$.

Invertible $\Rightarrow$ Full Rank

Suppose $XY = I_n$. Cross-fill $X = UV$ with $\text{rank}(X) = k$.

Then $\underbrace{U}_{n\times k}\underbrace{(VY)}_{k\times n} = I_n$. Recall §5's three cases:



- $k < n$: §5 says this **cannot** produce I_n — ruled out.
- $k > n$: But $k = \text{rank}(X) \leq n$ for an $n \times n$ matrix — ruled out.
- $k = n$: The **only** surviving case. $\implies$ $\text{rank}(X) = n$. ✓

Full Rank $\Rightarrow$ Invertible (Part 1: U is invertible)

($\Leftarrow$) Full rank $\Rightarrow$ invertible:

Suppose $\text{rank}(X) = n$. Cross-fill $X = UV$ (both $n \times n$).

Step 1: Cross-fill the augmented matrix $(X \mid I_n)$ with the **same pivots**:

$$(X \mid I_n) = U(V \mid W)$$

Comparing: $X = UV$ and $I_n = UW$.

Step 2: $UW = I_n$ (both $n \times n$), so by §5 Corollary: $WU = I_n$.

Therefore U is invertible with $U^{-1} = W$.

Full Rank $\Rightarrow$ Invertible (Part 2: V is invertible)

Step 3: Cross-fill the stacked matrix with the same pivots:

$$\begin{pmatrix} X \\ I_n \end{pmatrix} = \begin{pmatrix} U \\ Z \end{pmatrix} V$$

Comparing: $X = UV$ and $I_n = ZV$.

$ZV = I_n$ (both $n \times n$), so by §5 Corollary: $VZ = I_n$.

Therefore V is invertible with $V^{-1} = Z$.

Step 4: $X = UV$ with both invertible, so $X^{-1} = V^{-1}U^{-1}$. $\square$

Summary

What We Proved Today

1. Cross-filling projections (§1–§2): Each rank-1 piece is a projection; $P = UV$ with $P^2 = P$ forces $VU = I$.

2. Rank = Trace (§3): For any projection: $\text{rank}(P) = \text{trace}(P)$.

3. Orthogonal projections (§4): Diagonal cross-filling produces orthonormal bases of $\text{Col}(P)$ and $\text{Null}(P)$.

4. $UV = I \Rightarrow VU$ is a projection (§5): Three cases by shape — impossible ($r < n$), identity ($r = n$), genuine projection ($r > n$).

5. One-sided $\Rightarrow$ two-sided (§5): $UV = I_n$ with square matrices forces $VU = I_n$.

6. Full Rank $\iff$ Invertible (§6): An $n \times n$ matrix is invertible iff rank $= n$.

Preview: Lecture 9

Lecture 9: Compatible Projections

Today we proved that **cross-filling** a projection produces mutually annihilating rank-1 projections.

Does this generalize?

Suppose we write $P = R_1 + \cdots + R_k$ (not necessarily from cross-filling), with $\text{rank}(R_1) + \cdots + \text{rank}(R_k) \leq \text{rank}(P)$.

Must each R_i still be a projection? Must $R_i R_j = 0$?

The answer is **yes** — this is the **projection decomposition theorem**.